Chambers, universal constructions, and realization cones

Human Compression project

Abstract

We organize the mathematical content of Schulte and Ivić Weiss's 2006 survey *Problems on Polytopes, Their Groups, and Realizations*. Three common arguments treat distinguished generators and closed flag walks, universal amalgamation with regular local types, and finite equivariant realizations. Two oriented amalgamations are verified by their complete presentations. Previously established group results answer the sporadic existence question and the regular alternating rank question within their precise scopes. The source's remaining classification inputs and questions are recorded separately. In particular, its two aggregate counts for locally projective polytopes are reported claims, not reproduced censuses. All four component manuscripts form part of the proof; this synthesis does not replace their proofs by numerical summaries.

1 The source and the mathematical interfaces

The fixed source is arXiv:math/0608397v1, dated 15 August 2006, corresponding to the paper in *Periodica Mathematica Hungarica* 53 (2006), 231–255. We use its twenty-five physical pages and its complete TeX source. A later compilation date on the acquired PDF does not constitute a later mathematical edition. This is a survey with thirty-four numbered problems, rather than a paper with a single new classification theorem. Fourteen additional records separate embedded conjectures, restrictions, and unnumbered research directions. They are not fourteen additional independent conjectures.

An abstract polytope of rank n is a ranked poset with unique least and greatest faces, the diamond condition, and strong flag connectedness. The diamond condition says that each section of rank two has exactly two intermediate faces. Strong flag connectedness requires a path between two flags that retains their common faces. These conditions concern actual flags and sections; they cannot be replaced by connectedness of an unrelated incidence graph.

For a regular polytope, its distinguished involutions $\rho_0, \ldots, \rho_{n-1}$ satisfy the string commutations and

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_j : j \in J \rangle = \langle \rho_k : k \in I \cap J \rangle.$$

For a chiral polytope, adjacent flags are in the two different automorphism orbits. Its distinguished rotations $\sigma_1, \ldots, \sigma_{n-1}$ satisfy the string rotation relations and their rotation intersection condition. We retain the standard marked reconstruction theorems, attributed in the survey to McMullen–Schulte and Schulte–Weiss. Relations without the intersection condition do not supply a polytope. The geometric reconstruction is an explicit antecedent of the arguments below.

The complete general proof for highly symmetric hypertopes supplies the corresponding stabilizer argument. Paths within a residue identify its pointwise stabilizer with the subgroup on its remaining types. In the chiral case even paths give the pair-difference subgroup. Polytope flags are chambers of their face geometry, so the argument applies here. Conversely, a general C-group or C^+-group needs the geometric hypotheses stated in that proof. We use the marked string reconstruction where appropriate, not an unrestricted converse for arbitrary diagrams.

2 Words, orientations, and universal objects

Theorem 1. *For a regular or chiral polytope of rank $n \geq 3$, put $\alpha_0 = 1$ and $\alpha_i = \sigma_1 \cdots \sigma_i$ for $1 \leq i \leq n-1$. Simultaneous inversion of the α_i is equivalent to the usual reflection substitution*

$$\sigma_1 \longmapsto \sigma_1^{-1}, \qquad \sigma_2 \longmapsto \sigma_1^2 \sigma_2, \qquad \sigma_i \longmapsto \sigma_i \quad (i \geq 3).$$

For a chiral polyhedron, let s_1, s_2 denote its two distinguished rotations with the traversal convention in the component proof. The length of a j-hole is

$$\big|s_1 s_2^{1-j}\big|,$$

and the length of a j-zigzag is

$$2\big|s_1 s_2^{-j} s_1^{-1} s_2^{j}\big|.$$

Here $|g|$ denotes group order, and the traversal state includes the selected side of an edge.

Proof. The complete three-page component *Rotation conventions and closed flag walks* proves these statements by word cancellation and actual flag actions. The relevant identities are $\sigma_i = \alpha_{i-1}^{-1}\alpha_i$ and $\alpha_i^2 = 1$ for $i \geq 2$. If r_i is the operation of changing the type-i face of a flag, the traversal words are

$$h_j = r_0 r_1 (r_2 r_1)^{j-1}, \qquad z_j = r_0 (r_1 r_2)^j.$$

These r_i are flag permutations, not generally polytope automorphisms. An automorphism commutes with every adjacency operation. Consequently its iterates agree with successive repetitions of the chosen word on a base flag whenever one traversal returns to its automorphism orbit. The free action on that orbit identifies the traversal period with its group order. In a chiral action h_j preserves the orbit whereas z_j exchanges the two orbits. The component computes $h_j = s_1 s_2^{1-j}$ and $z_j^2 = s_1 s_2^{-j} s_1^{-1} s_2^j$ on the base flag. A return under z_j must therefore take an even number of steps, which gives the second formula. The same component gives the regular formulas and all convention changes explicitly. □

The formulas determine periods of a specified polyhedron. They do not determine which period lists determine a polyhedron, or when a universal quotient prescribed by those lists is finite. Those are the separate assertions in Problem 7.

Theorem 2. *Let K, L be regular rank-n polytopes, $n \geq 2$, with compatible distinguished common section. If any rank-$(n+1)$ polytope has facets K and vertex figures L, there is a regular universal polytope with these local types.*

Proof. The full two-page component *Universal regular amalgamation from a witness* gives the argument, including the group-action justifications. Start with involutions on the rank-$(n+1)$ string, imposing remote commutation. Add the complete marked relations of K on the first n generators and of L on the last n. In a given witness the adjacency permutations satisfy these relations at every flag: the local constituents are regular, so their marked relations hold throughout each local flag set. The presented group therefore acts on all witness flags.

The two end subgroups identify with the exact local groups. Surjections in one direction come from the local presentations; their actions on the corresponding local flags give the reverse order-free identifications. An element in both end subgroups preserves the base facet and vertex. Strong flag connectedness of their common section supplies a middle-subgroup element with the same effect on the base flag. The free action of either end group makes these two elements equal. Thus the end intersection is exactly the middle subgroup. The retained string end criterion and reconstruction theorem give a regular polytope. Repeating the action on every other witness supplies the covering map from this one, and proves universality. □

The witness assumption is essential. The source already gives a pair with no regular amalgamation: facets $\{4,4\}_{(3,0)}$ and hemicube vertex figures. That negative example is retained with its attribution to *Abstract Regular Polytopes*, p. 367. Theorem 2 supplies neither existence for arbitrary compatible pairs nor a finiteness criterion. The analogous theorem for oriented chiral local constituents is a separate retained Schulte–Weiss input. It requires the chosen enantiomorphic forms and directly regular constituents where a constituent is regular.

Proposition 3. *The two oriented universal amalgamations with local pairs*

$$(\{4,4\}_{(1,3)},\{4,4\}_{(1,3)}) \quad \textit{and} \quad (\{4,4\}_{(3,1)},\{4,4\}_{(1,3)})$$

are chiral and have full automorphism groups of orders 2000 *and* 960*, respectively.*

Proof. The complete two-page component *Two oriented square-torus amalgamations* verifies the full presentations

$$\langle u,v,w \mid u^4=v^4=w^4=(uv)^2=(vw)^2=(uvw)^2=1,\\ (u^{-1}v)^s(uv^{-1})^t=(v^{-1}w)(vw^{-1})^3=1\rangle$$

at $(s,t)=(1,3),(3,1)$. Exact isomorphisms from the finitely presented groups to permutation groups give the stated orders. The two end groups have order 40 and their intersection is $\langle v\rangle$ of order 4; both adjacent cyclic intersections are trivial. Separate local presentations also have order 40, so the local types do not collapse. Both toroidal relators fail after the reflection substitution of Theorem 1. The retained rotation reconstruction therefore gives chirality. The presentation imposes exactly the local relations, so its universal property is the retained oriented amalgamation correspondence. Independent literal permutation closure checks the finite orders and intersections; the upper order of the presentation itself remains the explicit GAP isomorphism calculation. These established examples are attributed by the survey to Hubard–Weiss (2005), p. 132. □

3 Finite equivariant realization theory

Let a finite group G act on a nonempty finite set X. A realization is a map $f: X \to V$ to a real orthogonal G-module with $f(gx)=gf(x)$. We allow repeated vertices until faithfulness is imposed, translate to centroid zero, and take the actual linear span of the image. Write $\mathbf{1}$ for the all-ones vector and $J=I-|X|^{-1}\mathbf{1}\mathbf{1}^T$.

Theorem 4. *Congruence classes of centered equivariant realizations are identified with*

$$\mathcal{C}_G(X)=\{B=B^T\succeq 0 : B\mathbf{1}=0,\ gB=Bg \text{ for every } g\in G\},$$

where g acts by its permutation matrix on $\mathbb{R}^X$. The dimension of the realized span is $\operatorname{rank} B$*. Squared distances and Gram matrices are mutually recoverable by*

$$D_{xy}=B_{xx}+B_{yy}-2B_{xy}, \qquad B=-\tfrac{1}{2}JDJ.$$

Blending realizations is addition in this cone. A nonzero B spans an extreme ray precisely when $\operatorname{im} B$ *is an irreducible real G-module.*

Proof. The complete four-page component *Finite realization cones* proves the Gram correspondence, the inverse on hollow symmetric distance matrices, and the extreme-ray statement. The realization of B is $f_x=B^{1/2}e_x$ in its actual image. Orthogonal direct sums give addition of Gram matrices; one then restricts to the span rather than assuming that the entire direct sum is spanned.

For clarity, the point where irreducibility enters is as follows. If $B = C + D$ with $C, D \succeq 0$, then $\ker B \subseteq \ker C$, so C restricts to $\operatorname{im} B$. An invariant self-adjoint operator on an irreducible real module is scalar: each of its real eigenspaces is invariant. This applies to both B and C, and makes C a scalar multiple of B. Conversely, if $\operatorname{im} B$ is reducible, choose an orthogonal invariant projection Q onto a nonzero proper summand. With $A = B^{1/2}$ the decomposition

$$B = AQA + A(I - Q)A$$

has two nonzero invariant positive semidefinite terms. Their ranks preclude either being a multiple of B. This argument does not assume that Q commutes with B. □

For a transitive vertex set $X = G/H$, all such orbit maps have the form $gH \mapsto gv$ with $v \in V^H$. Distinct abstract vertices have distinct images exactly when $\mathrm{Stab}_G(v) = H$. Faithfulness of the representation alone does not imply this equality. For a polytope one must also separate distinct faces of each rank under the recursive face construction. The component gives the precise recursive test, and identifies the full geometric symmetry group as the Gram stabilizer in the abstract automorphism group, acting on the actual affine span.

The finite-group assumption is used, in particular, in averaging and in the orthogonal representation setting. The source's infinite Euclidean Wythoff constructions, discreteness conditions, and geometric classifications are retained separately. Theorem 4 gives neither all individual toroid cones nor the answer about quaternionic irreducible characters in Problem 23. It also does not decide whether an abstract regular polytope has a faithful chiral geometric realization. The realization theory retains the attribution to McMullen and to McMullen–Monson recorded in the source; a new proof of its common linear algebra is not a claim of priority for that theory.

4 Group consequences already available

The acquired simple-group theorem of Leemans–Liebeck (2016) constructs a chiral polyhedron for every nonabelian finite simple group outside A_7, $\mathrm{PSL}_2(q)$, $\mathrm{PSL}_3(q)$, and $\mathrm{PSU}_3(q)$. We use its positive assertion only. All twenty-six sporadic groups lie in its scope, including the Monster, so it answers the existence request in Problem 31. This is a previously established answer, with its original attribution, not a new solution obtained here or an enumeration of all possible polytopes with those groups.

The acquired alternating-group existence and upper-rank theorems, together with their complete finite boundary censuses, give the following regular ranks for $n \geq 3$:

n	ranks of string C-markings generating A_n
$3, 4, 6, 7, 8$	none
5	3
9	$3, 4$
10	$3, 4, 5$
11	$3, 6$
$n \geq 12$	$3, 4, \ldots, \lfloor (n-1)/2 \rfloor$.

The exceptional gaps at $n = 11$ are retained from the complete finite census; they are not filled by interpolation. The lower existence theorem and upper bound have separate source attributions in their acquired manuscripts. The simple-group theorem also supplies chiral rank-three examples for A_n, $n \geq 8$. These statements do not classify all chiral ranks or all other polytopes requested in Problem 32.

For Problem 30, the acquired common quotient for a nondegenerate finite string C-marking of a 2-group in rank $d \geq 2$ has order 2^{2d-1} and nilpotency class at most two. The eight acquired families with large adjacent generator orders have their stated parameter hypotheses. Neither result classifies every 2-group, every chiral marking, or the groups of order $2^k p$ in the source question.

Proposition 5. *The full automorphism group of a finite neighborly regular or chiral polytope of positive rank acts* 2*-transitively on its vertices.*

Proof. First suppose the rank is at least three. Given an ordered pair (v, w) of distinct vertices, neighborliness supplies an edge e with these endpoints. Extend the incident vertex-edge pair (v, e) to a flag. In the regular case all flags lie in one orbit. In the chiral case, if the flag is in the unwanted orbit, change only its type-2 face. This preserves v and e and moves the flag into the chosen orbit. Do the same for any second ordered vertex pair (v', w'). An automorphism taking the resulting first flag to the second takes v to v' and e to e'. An edge has exactly two vertices by the diamond condition, so it also takes w to w'. Thus all ordered pairs of distinct vertices are in one orbit. In rank one the two vertices are interchanged. In rank two a neighborly polygon has two or three vertices, and its dihedral action is 2-transitive on them. Chiral polytopes have rank at least three by definition. □

This proves the necessary condition stated after Problem 33. It is not a classification or a converse, and is not claimed as a new theorem in the subject.

5 Retained classifications and precise limits

The survey collects results from many different sources. The accompanying register has fifty-two clauses in sixteen contexts, with source pages, literal anchors, citation keys, and each acquired proof or retained premise. Its sixty-six bibliography entries preserve the complete source list. The following table describes the division of work. A retained classification is an attributed input, not a classification reproved by the arguments above.

Context and register	Proof or retained input
C01: Definitions and chambers R01	Adopt the definitions and retain the marked reconstruction theorems. The acquired residue proof supplies the stabilizer identities under actual polytope hypotheses.
C02: Orientations R02	The complete word proof identifies both reflection conventions. Index at most two for rotations does not mean index exactly two.
C03: Dualities and maps R03–R08	Retain the Archdeacon–Richter and Servatius sphere results, Hubard–Weiss polarity theorem, Jendrol counterexample, and attributed toroidal and genus statements. The extra fully transitive toroidal family is not chiral.
C04: Hole and zigzag periods R09	The full flag-walk proof gives all stated periods. It gives no classification from period data.
C05: Amalgamation R10–R16	Prove the regular universal theorem from a witness and reproduce the two oriented examples. Retain oriented universality, the ARP cross-polytope criterion, the negative pair, and historical extension/rank-five assertions.
C06: Global topology R17–R19	Retain the spherical, projective and toroidal classifications and chiral restrictions from Coxeter and ARP, with their tessellation hypotheses.
C07: Local spherical and projective types R20–R23,R52	Retain the universal Coxeter tessellation theorem and the reported finiteness distinction between projective ranks four and five. Acquire the exact J1 times PSL2(19) example and its corrected family census. The global17/441 aggregates remain unverified source reports.
C08: Local toroidal types R24–R29	Retain the precise rank/type and regular finite-family classifications in ARP. Conditional regular universality explains the rank-five/six universal restriction. Finite-list conjectures remain separate.
C09: Other local surfaces R30–R31	The four orientability combinations are definitions. Retain the historical catalogue report with its unverified chiral-citation scope.
C10: Equivariant realizations R32,R50	Prove the finite orbit-stabilizer and recursive faithfulness criteria. Infinite Euclidean Wythoff constructions remain separate retained inputs with discreteness and exact stabilizers required.

Context and register	Proof or retained input
C11: Realization cones	Prove the common centered Gram-cone and irreducibility theorems.
R33–R34	Individual cone catalogues and unevaluated character identities retain their published scope.
C12: Geometric classifications	Retain Grunbaum–Dress, Schulte, McMullen, and the finite planar-faced uniform classifications. Skew-faced and infinite cases are not covered by the
R35–R38	finite planar list.
C13: Medial graphs	Retain the precise rank-four type{3,q,3} results and their arc-transitivity or
R39–R40	semisymmetry hypotheses. The reverse construction has a separate polytopality condition.
C14: Petrie schemes	Retain the attributed positive geometric classes and necessary local
R41–R42	condition. A return of a full flag does not exclude repetition of a proper face.
C15: Specified groups	Acquire the complete sporadic existence and regular alternating ranks, and
R43–R46,R51	only the specified restricted2-group results. Prove the neighborliness implication, with no classification converse.
C16: Quotients	Retain Hartley quotient and semisparseness theorems with every conjugate
R47–R49	quantifier. The general end-condition converse remains a distinct question.

The source also reports that every locally projective regular polytope of rank four is finite, whereas this assertion fails in rank five. We retain this distinction as a classification input attributed to Hartley and to *Abstract Regular Polytopes*. The full primary classification has not been independently inspected; the common proofs above do not establish this assertion.

There are two numerical limits of particular importance. The source reports seventeen universal locally projective regular 4-polytopes, eight with a 2 in their Schläfli symbol, and 441 quotients including four additional nonuniversal regular polytopes. These are attributed to Hartley's then-preprint classification. We have not reproduced the global lists or reconciled the complete preprint and published editions. They therefore remain reported aggregates, not verified conclusions of this paper.

One constituent is acquired in full: the universal $\{\{5,3\},\{3,5\}_5\}$ has group $J_1 \times \mathrm{PSL}_2(19)$, and its accepted corrected quotient census has 137 entries under its specified equivalence. It would be invalid to subtract a correction for this family from 441 without the original global partition, overlap conventions, and edition correspondence. No corrected global total is asserted.

The source's report of map enumerations through orientable genus 15 and nonorientable genus 30 includes a chiral claim whose cited paper is titled *Determination of all regular maps of small genus*. The inspected material does not independently supply that chiral census. We retain the historical report with this explicit limitation.

The source's semisparseness criterion is likewise retained with its actual quantifiers. Put $\Gamma_{<i} = \langle \rho_j : j < i \rangle$, $\Gamma_{>i} = \langle \rho_j : j > i \rangle$ and $\Gamma_i = \langle \rho_j : j \neq i \rangle$. For $\Sigma \leq \Gamma$, the first two tests are

$$\begin{aligned} \Sigma^\varphi \cap \Gamma_i \rho_i &= \varnothing, \\ \Gamma_{>i} \cap \Gamma_{<j} \Sigma^\varphi &\subseteq \Gamma_k (\Gamma_{>i} \cap \Gamma_{<j}) \Sigma^\varphi, \qquad i < k < j, \end{aligned}$$

for every $\varphi \in \Gamma$ and all displayed valid indices. The third is written in the source as

$$\Gamma_{>i} \cap \Gamma_{<i+1} \Sigma \subseteq \Sigma \quad (0 \leq i < n-1).$$

Hartley's further necessary condition requires both end intersections to be semisparse for every conjugate and

$$\Sigma^\varphi \cap \Gamma_0 \Gamma_{n-1} = (\Sigma^\varphi \cap \Gamma_0)(\Sigma^\varphi \cap \Gamma_{n-1}) \quad (\varphi \in \Gamma).$$

Problem 34 asks for sufficiency of these latter conditions. The special locally projective census, with its additional hypotheses, does not prove that general converse.

6 The questions and the fixed edition

The table below records every numbered problem. “Unsettled here” states a limit of the present acquired proofs; it does not assert that the question is still open in the current literature. A broader search for later answers is outside this intake. The fourteen secondary records follow the table so that an embedded conjecture cannot disappear within a broader programme.

No.	Question in the fixed edition	Disposition and qualification
1	Classify self-dual toroidal polyhedra.	Unsettled here. The spherical classification is attributed background; regular or chiral toroidal maps are only a subclass.
2	Decide whether every equivelar self-dual polyhedron has a polarity.	Unsettled here. The source gives the affirmative regular/chiral subclass and a negative unrestricted self-duality example; neither answers the equivelar question.
3	Classify fully transitive polyhedra on surfaces of small genus, with both orientability choices.	Unsettled here. The toroidal classification cited in the source includes a further family whose middle adjacency preserves each flag orbit.
4	Determine the least orientable genus at least two admitting a nonregular fully transitive polyhedron.	Unsettled here. The source excludes chiral maps in genera two through six; fully transitive is a larger class.
5	Characterize groups of fully transitive polyhedra outside the regular and chiral classes.	Unsettled here. The subsequent self-dual restriction is an additional suggested subproblem.
6	Classify edge-transitive polyhedra on orientable and nonorientable surfaces of small genus.	Unsettled here. The cited partial results do not give the requested classification.
7	Determine what all specified hole and zigzag lengths determine, and when the associated universal regular quotient is finite.	Unsettled here. The first question covers regular or chiral polyhedra of type {p,q}, with 1¡=j¡=floor(q/2). The universal regular quotient in the subsequent question is distinct. Word-order formulas alone give neither uniqueness nor a universal finiteness criterion.
8	Decide existence with prescribed regular or chiral facets and vertex figures.	Already negative in the source. This is already negative in general in the source: the cited toroidal (3,0) facet and hemicube vertex figure have no regular amalgamation.
9	Characterize the compatible facet and vertex-figure pairs for which amalgamation exists.	Unsettled here. Matching their common sections is necessary but insufficient.
10	Determine universal groups and their finiteness for regular facet and vertex-figure pairs admitting amalgamation.	Unsettled here. The source existence theorem for the universal object is retained separately from group classification.
11	Determine universal groups and finiteness for oriented chiral/regular amalgamations with at least one chiral member.	Unsettled here. Choose enantiomorphic forms, with at least one chiral constituent and every regular constituent directly regular. Simultaneous reversal identifies four oriented choices in pairs; if one constituent is regular the remaining choices also agree. Existence remains a prerequisite.
12	Determine existence and finiteness with an arbitrary finite regular facet and a convex regular vertex figure.	Unsettled here. The source records the cross-polytope vertex-figure case: finiteness is equivalent to neighborliness of the facet.
13	Determine existence and finiteness with finite chiral facets and convex regular vertex figures.	Unsettled here. The regular cross-polytope result does not automatically extend to chiral facets.

No.	Question in the fixed edition	Disposition and qualification
14	Establish existence of finite chiral polytopes in ranks at least six.	Unsettled here. Affirmative answer conjectured in the 2006 source. This inventory makes no present-day open-status assertion.
15	Decide whether a convex four-polytope can have every facet combinatorially icosahedral.	Unsettled here. An abstract polytope or incidence geometry with such facets is not sufficient.
16	Classify universal regular amalgamations of square toroidal maps with odd distinct parameters s,u at least three.	Unsettled here. Two claims are embedded: existence for every such pair and finiteness only for (3,5) and (5,3).
17	Classify universal regular amalgamations of triangular and hexagonal toroidal maps in the stated regular parameter families.	Unsettled here. Retain the exact source parameter domains: s¿=2,t=0 or s=t¿=1, and the corresponding u,v alternatives.
18	Complete the classification of the three locally toroidal universal regular rank-six families, up to duality.	Unsettled here. The completeness of each finite list and the explicit one-relation infinite-family conjecture are separate clauses.
19	Classify the universal locally toroidal chiral rank-four polytopes with the specified regular or chiral constituents and enantiomorphic forms.	Unsettled here. The preceding discussion and subsequent group calculation focus on the finite classification; retain that goal separately from the literal universal-object wording. At least one constituent is a chiral toroidal map. The square-torus/cube subfamily has its own four-pair finiteness conjecture.
20	Classify finite universal regular or chiral rank-four polytopes with prescribed small facet and vertex-figure genera and orientabilities.	Unsettled here. Duality exchanges the two mixed orientability cases; the problem is not restricted to locally toroidal polytopes.
21	Develop the realization theory of chiral polytopes.	Unsettled here. A theory of equivariant Gram matrices alone does not classify faithful realizations or determine their full geometric symmetry.
22	Describe realization cones of regular toroids in every rank.	Unsettled here. The cited rank-three cones are established background, not the requested all-rank description.
23	Decide whether a finite string C-group can have a complex irreducible character with Frobenius-Schur indicator minus one.	Unsettled here. The representation must be quaternionic; a general real representation or a non-C-group example is insufficient.
24	Determine intersection-property validity from character tables or other readily computed group-algebra invariants for a marked string-generated group.	Unsettled here. The marking is part of the input. An explicit subgroup enumeration is not by itself an answer to the character-data question.
25	Enumerate discrete fully transitive polyhedra in Euclidean three-space, both finite and infinite.	Unsettled here. Regular and chiral classifications supply subclasses only.
26	Enumerate discrete uniform polyhedra in Euclidean three-space, both finite and infinite.	Unsettled here. The finite planar-faced list does not include all skew-faced or infinite cases.
27	Characterize when the source construction from a finite bipartite trivalent two- or three-arc-transitive graph yields a polytope.	Unsettled here. The condition must concern the actual reconstructed poset; graph transitivity alone is not asserted sufficient.

No.	Question in the fixed edition	Disposition and qualification
28	Decide whether all Petrie schemes are acoptic for each of four specified classes of products, uniform convex polytopes and regular star-polytopes.	Unsettled here. The four classes are: products of cubes and crosspolytopes; products of cubes or crosspolytopes and 4k-gons (k¿=1); uniform convex polytopes; regular star-polytopes. Affirmative answers are conjectured for classes 1,2,4; class 3 remains a separate question.
29	Characterize universal regular amalgamations for which every Petrie scheme is acoptic.	Unsettled here. The stated local nonintersection condition on constituents is only necessary.
30	Characterize full automorphism groups of regular or chiral polytopes of order a power of two or a power of two times an odd prime.	Restricted acquired results only; general question unsettled here. Existing project results cover specified ordered regular markings, bounded class/excess and exact small excesses; the unmarked, chiral and mixed-prime problem remains outside them.
31	Find polytopes with sporadic simple full automorphism groups, explicitly including the Monster.	Affirmative existence by the acquired 2016 theorem. The accepted positive theorem of the 2016 simple-group manuscript gives a chiral polyhedron for each of the26 sporadic groups. This is an acquired earlier answer, not a new conjecture proof.
32	Find regular, chiral, or other polytopes with full automorphism group A_n; in particular determine the possible degrees for each rank in the regular or chiral classes.	Regular ranks acquired; chiral rank-three existence only; other parts unsettled here. For n¿=3, acquired L038/L044 and their complete small censuses settle the regular ranks: A5:{3}; A9:{3,4}; A10:{3,4,5}; A11:{3,6}; none at n=3,4,6,7,8; and exactly 3¡=r¡=floor((n-1)/2) for n¿=12. Acquired L045 gives chiral rank-three existence for n¿=8. This does not classify all chiral ranks or all other polytopes. These are attributed earlier results, not newly solved questions.
33	Classify finite neighborly regular or chiral polytopes.	Unsettled here. The necessary two-transitive vertex action is not asserted sufficient for reconstruction.
34	Decide whether semisparsity follows from semisparsity of both end intersections in every conjugate and the displayed product-intersection equality.	Unsettled here. Retain every conjugate quantifier; necessity and the general three-condition definition do not prove this sufficiency assertion.

Secondary records.

Record	Exact scope of the additional record
Problem 5: selfdual	Seek the group characterization first for self-dual fully transitive polyhedra. No new solution supplied.
Problem 14: existence	Affirmative finite chiral existence conjecture in every rank at least six. No new solution supplied.
Problem 16: existence	Existence for all odd distinct s,u¿=3. No new solution supplied.
Problem 16: finiteness	Finite precisely at (3,5) or its dual. No new solution supplied.
Problem 18: lists	The known finite lists in all three rank-six families are complete. No new solution supplied.
Problem 18: first family	The specified Coxeter quotient is an infinite C-group outside (2,0,0,0),(2,2,0,0),(3,0,0,0). No new solution supplied.
Problem 19: cube	The square-torus/cube universal chiral group is finite precisely at (1,2),(1,3),(1,4),(2,3), up to transposition. No new solution supplied.

Record	Exact scope of the additional record
faithful chiral	Can an abstract regular polytope admit a faithful chiral geometric realization? No new solution supplied.
abstract uniform	Define abstract uniform polytopes and develop their abstract and geometric theories. No new solution supplied.
helices	Determine general properties of polyhedra with helical faces. No new solution supplied.
Problem 28: three classes	All Petrie schemes are acoptic in classes1,2,4. No new solution supplied.
Problem 31: finite simple	Corresponding existence question for arbitrary finite simple groups, with the historical regular-map question distinguished. Only the exact positive families of the acquired simple-group theorem are supplied; the excluded families and regular-map question are not decided.
star vertex figures	Extend the existence/finiteness questions by allowing a regular star-polytope as vertex-figure. No new solution supplied.
prescribed facets	Find and describe regular or chiral polytopes with a prescribed facet isomorphism type, including variants with a prescribed last Schlaefli entry. No new solution supplied.

Several edition qualifications matter for interpreting these statements. Within the nondegenerate fundamental domain used in the source, the chiral toroid condition is $t \neq 0$ *and* $s \neq t$; the printed "or" would include the regular cases. This correction alone does not prove that every small parameter gives a nondegenerate polytope. The reference to plane maps at rank $n = 2$ is inconsistent with the source's rank-three polyhedron convention and is retained as an edition qualification. Recursive face realization is used at all appropriate ranks, rather than extrapolating the displayed $j = 1, 2, 3$ notation without a definition. The abstract and introduction give different meeting years; this historical discrepancy does not change the fixed mathematical edition. Three repeated TeX labels refer to different numbered problems, so our register identifies problems by ordinal number and source position, never by the duplicated label alone.

7 Proof record and attribution

The three common theoretical components contain complete proofs: four pages for finite realization cones, three for rotation conventions and flag walks, and two for conditional regular amalgamation. The two finite examples add two pages with complete presentations, executable GAP calculations, and independent literal permutation checks. All four have distinct reviews. Their unabridged texts are appended to the reading volume. The earlier group and locally projective results are acquired with their full proof and certificate dependencies, not just their final orders or page counts.

The new work here is the arrangement of these proofs, the explicit joins between their hypotheses, and the short proof of Proposition 5. The source authors and their cited predecessors retain attribution for established statements. The source register separately identifies definitions, retained classifications, historical reports, acquired proofs, and questions. No new open-problem solution is claimed. The computational checks are not Lean certificates, and no new compression factor is inferred from this survey disposition. Its admission as a handled original requires the separate review of this entire assembly.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on Polytopes, Their Groups, and Realizations*, Period. Math. Hungar. 53 (2006), 231–255; arXiv:math/0608397v1. Fixed source and complete bibliography.

[2] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002. Standard reconstruction and the classifications at the locations specified in the retained-input register.

[3] E. Schulte and A. Ivić Weiss, *Chirality and projective linear groups*, Discrete Math. 131 (1994), 221–261. Retained through the inspected survey, not independently reacquired here.

[4] I. Hubard and A. Ivić Weiss, *Self-duality of chiral polytopes*, J. Combin. Theory Ser. A 111 (2005), 128–136. Attribution and oriented examples as cited in the fixed source.

[5] Human Compression project, *Finite realization cones*; *Rotation conventions and closed flag walks*; *Universal regular amalgamation from a witness*; *Two oriented square-torus amalgamations*. Complete acquired manuscripts and distinct reviews.

[6] The exact acquired group, chamber, and locally projective manuscripts are listed with their source attributions and immutable editions in the acquired-proof register. Their original results receive no new priority or question credit in this paper.